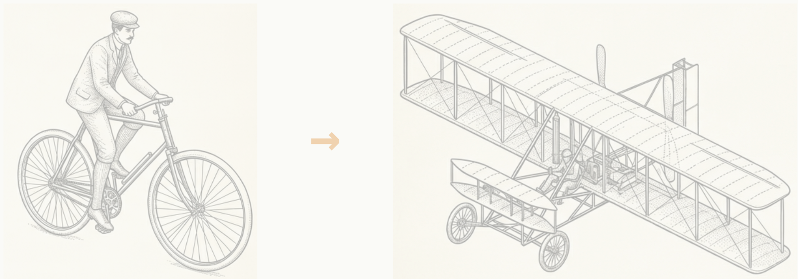
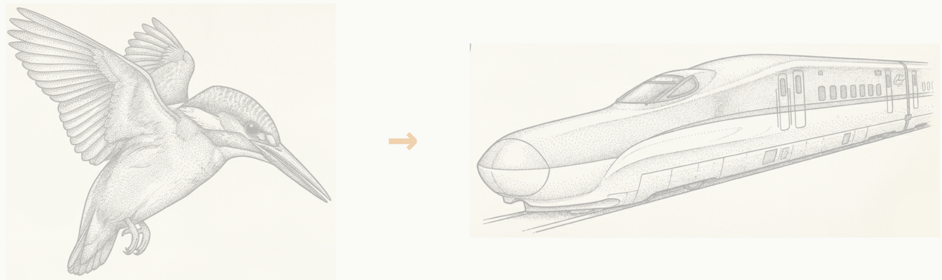
1903 FIG. 01



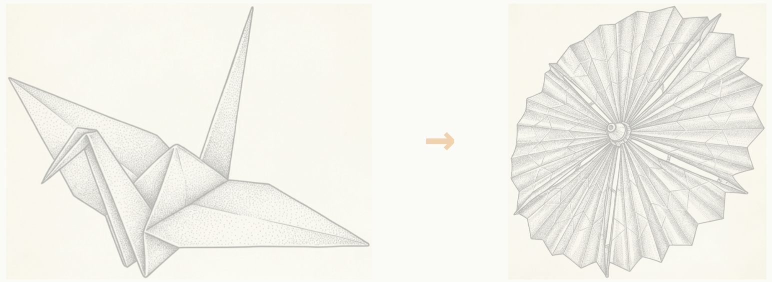
BICYCLE → FLYING MACHINE

1997 FIG. 02



KINGFISHER BEAK → SHINKANSEN NOSE

2016 FIG. 03



ORIGAMI → SOLAR SAIL

Carnegie Mellon University | HCII Human-Computer Interaction Institute

KITTUR LAB · HUMAN-COMPUTER INTERACTION INSTITUTE

Solving industry’s hardest problems with solutions from *unexpected places*.

What the research offers, which companies to bring us, and how it can help you open doors.

Niki Kittur · Professor, HCII

FOR CMU CORPORATE RELATIONS · SEPTEMBER 2026

CONFIDENTIAL

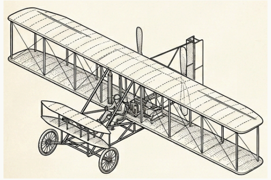
THE PROBLEM EVERY COMPANY BRINGS

Every R&D team is stuck on problems they’re making only *incremental progress* on.

Because experts search where they already know — and AI gets you more of what’s already out there.

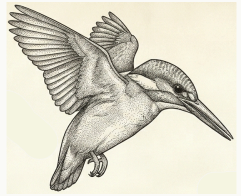
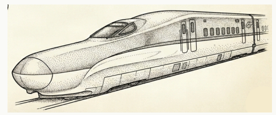
Our research makes finding breakthroughs from distant domains *systematic*: retrieve by mechanism, not keyword; map it into their domain; de-risk it into something their team can build and test.

BREAKTHROUGHS COME FROM ANALOGIES

 → 

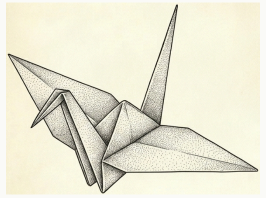
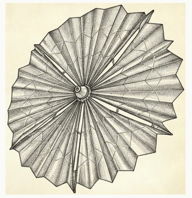
Bicycle → Wright Flyer (1903)

Frame-lean steering became wing-warp steering: the control problem that grounded every prior aviator.

 → 

Kingfisher → bullet train (1997)

A beak that enters water without a splash became a nose that ends sonic booms in tunnels.

 → 

Origami → NASA solar sail (2016)

A Miura fold packs a 25 m sail into a 1 m fairing: a packaging problem became a paper-folding one.

Serendipity found these. We make them searchable — forward, and in reverse (a capability → the problems it could solve).

12+ years of research at HCII   *PNAS* · best papers at CHI, CSCW, KDD

NSF-funded · 4 patents   Kittur · Martelaro · Mohanty · Shahaf

WHAT IT LOOKS LIKE · 60 SECONDS

A public P&G challenge: *clean between teeth without floss.*

Nobody in oral care would look at an oil rig. The engine did – and came back with a concept, its evidence, and the tests to run.

The concept: food-safe iron-oxide microbeads (E172) self-assemble under a rotating field into spinning soft augers that scrub the gaps, then rinse away.

01 SEARCH

Retrieve by *mechanism*: “dislodge a stubborn deposit from a narrow channel.” Matches came from **drilling fluids** that scour a borehole and **vascular medicine**, where magnetic bead swarms clear clots.

02 TRANSFER

Map the mechanism onto the real constraints: safe in the mouth, no hardware a consumer won’t use, a sachet and a rinse. Every claim is tied to a primary source – not a hallucinated one.

03 EVALUATE

Hand back the four test protocols and success criteria that settle the risk: chain formation, biofilm softening, debris removal, bead recovery. The team knows what to run Monday.

The same loop runs on a materials scientist’s holy-grail problem or a propulsion chemist’s storage problem – with their experts in the loop.

PROOF · WITH INDUSTRY PARTNERS

Ideas their own experts rated as ones *they couldn't have reached*.

COVESTRO · FORTUNE 500 MATERIALS

3 "HOLY-GRAIL" PROBLEMS STUCK 10+ YEARS · 17 SCIENTISTS

54%

more novel, no drop in feasibility ($p < .05$)

30+

promising ideas vs. 0–1 from their own LLM tools

2

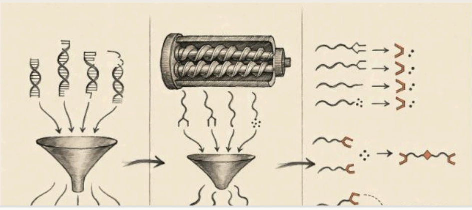
already in their managed innovation pipeline

IDEA 01 OF 03 NEXT ›

BORROWED FROM DNA-SEQUENCING PREPARATION

Convergent chain repair: boron funnels every broken end into one joinable handle

Borrow the terminus-funneling step from next-generation sequencing library preparation to converge every broken polyolefin chain end onto one boron handle and rebuild linear molecular weight without radicals or gel.



HOW IT WORKS

Thexylborane, carrying exactly two B–H bonds, hydroborates terminal vinyl and vinylidene chain ends in the melt, capturing two chains per molecule onto one dialkylborane handle; a solid-acid co-additive funnels carbonyl-derived ends into the same pathway via dehydration. Downstream CO injection drives Brown carbonylation [3], migrating both chains onto a carbonyl carbon to yield a linear ketone bridge – purely ionic, no radical generated, mid-chain attack impossible. No one has combined

"If we sat down and spent a week going through these things, that would be a worthwhile investment to make sure that we're spending the next two years on the right projects." — R&D group leader

URSA MAJOR · AEROSPACE PROPULSION "HOW CAN WE EXTEND THE LIFE OF OUR PROPELLANT?"

+80%

estimated propellant life at minimal extra cost vs. 5–10% from their previous approach

Paid pilot underway.

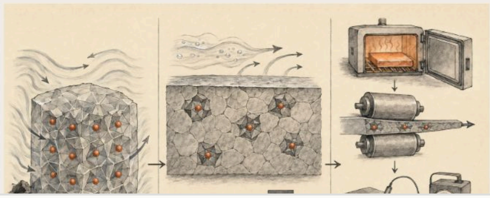
Concepts borrowed from semiconductor wafer gettering and nuclear-waste ceramics; scope expanding.

IDEA 03 OF 03 ‹ PREV

BORROWED FROM NUCLEAR-WASTE CERAMICS

Lattice-pinned alloy: heat treatment locks trace metals where peroxide cannot reach

Borrow lattice-site immobilization from nuclear-waste ceramics to trap catalytic trace metals in aluminum tank plate into crystal sites hydrogen peroxide physically cannot breach, enabling decade-scale storage without hazardous passivation.



HOW IT WORKS

Annealing 5254 plate at 480 °C drives residual Fe and Cu into Al_3Fe and $Al_6(Fe,Mn)$ intermetallics [26], [32] whose formation enthalpies exceed the oxidation potential H_2O_2 can deliver, locking catalytic metals in lattice sites the oxidizer cannot breach. Current practice controls only composition and surface chemistry [34]; this opens a third axis – microstructural site occupancy – set by a single upstream heat treatment before cold-rolling to flight strength. Intermetallics formed above 400 °C are

"If this was [available to every engineer] they would [use it] daily." "...ideas which no one in my team could come up with." — R&D leader, Fortune 500 automotive

BUILT ALONGSIDE Covestro · Ursa Major · Toyota · Bosch · Conservation X Labs · and a public P&G challenge

WHO TO BRING US

Industrial R&D with a *named, stuck problem* – or an underused capability.

partner · in conversation · plain: target

MATERIALS & CHEMICALS

Covestro · BASF · PPG · Dow · DuPont · 3M · Henkel · Solvay · Evonik · SABIC · Arkema

CONSUMER GOODS & FOOD

P&G · PepsiCo · Chobani · Unilever · Nestlé · Mondelez · Mars · L'Oréal · Kraft Heinz · Colgate-Palmolive

AEROSPACE & DEFENSE

Ursa Major · RTX · Lockheed Martin · Northrop Grumman · Boeing · Honeywell · Anduril · Blue Origin

AUTOMOTIVE & MOBILITY

Toyota · Bosch · GM · Ford · Stellantis · Honda · Magna · Cummins

ENERGY & INDUSTRIAL

Westinghouse · Wabtec · Alcoa · GE Vernova · Siemens Energy · SLB · Eaton · Caterpillar

ELECTRONICS & SEMICONDUCTORS

Mitsubishi Electric · Intel · Micron · Corning · Applied Materials · Samsung · TSMC

HOW TO SPOT A FIT

The right person, the signal, and *the question that opens it*.

01 THE RIGHT PERSON

VP or Head of R&D

CTO · Chief Science or Chief Innovation Officer

Open-innovation and alliance-management leads

The BD lead looking for new markets for an existing material or IP

Not the CIO, and not procurement — the person who owns the technical roadmap and the problems on it.

02 THE SIGNAL IT'S A FIT

A problem their team has been on for years

A paid-for capability whose best markets they can't see

They've tried the LLMs and been underwhelmed

An open-innovation mandate and appetite for a pilot

One of these is enough. Two and it's a strong lead.

03 THE QUESTION THAT OPENS IT

"What's the problem your best people have been on the longest?"

"What have you paid to develop that you suspect has uses you can't see from here?"

"Where did the AI tools your teams tried fall short?"

Send us the answer. We come back with directions and the CMU people who could pursue them.

HOW THIS CAN HELP YOU

Walk into the first meeting with three concrete directions – and the CMU people who could pursue them.

01 Start the conversation

A company's interest is nebulous, or they name a problem they've been stuck on. Send it to us before the first meeting. You walk in with three concrete directions on *their* problem and the faculty across CMU who could pursue each one – a match, not a brochure.

02 Sponsored research & workshops

Through the lab: a half-day ideation session with their R&D team on a named problem, or a scoped sponsored project on the thing they've been stuck on. Their experts stay in the loop; our students and postdocs do the work.

03 Route to the spinout

For deployment inside a company, the CMU spinout Analogical Engines productizes this research – and an NSF grant to commercialize it keeps the company tied to CMU for the next three years. Route each inquiry to the right door: exploratory work through the lab, deployment through the spinout.

BRING ME A COMPANY AND A PROBLEM

nkittur.github.io/unexpected-places

Case studies, what we offer, who to bring · nkittur@cs.cmu.edu

Happy to talk to any of you.

Niki Kittur